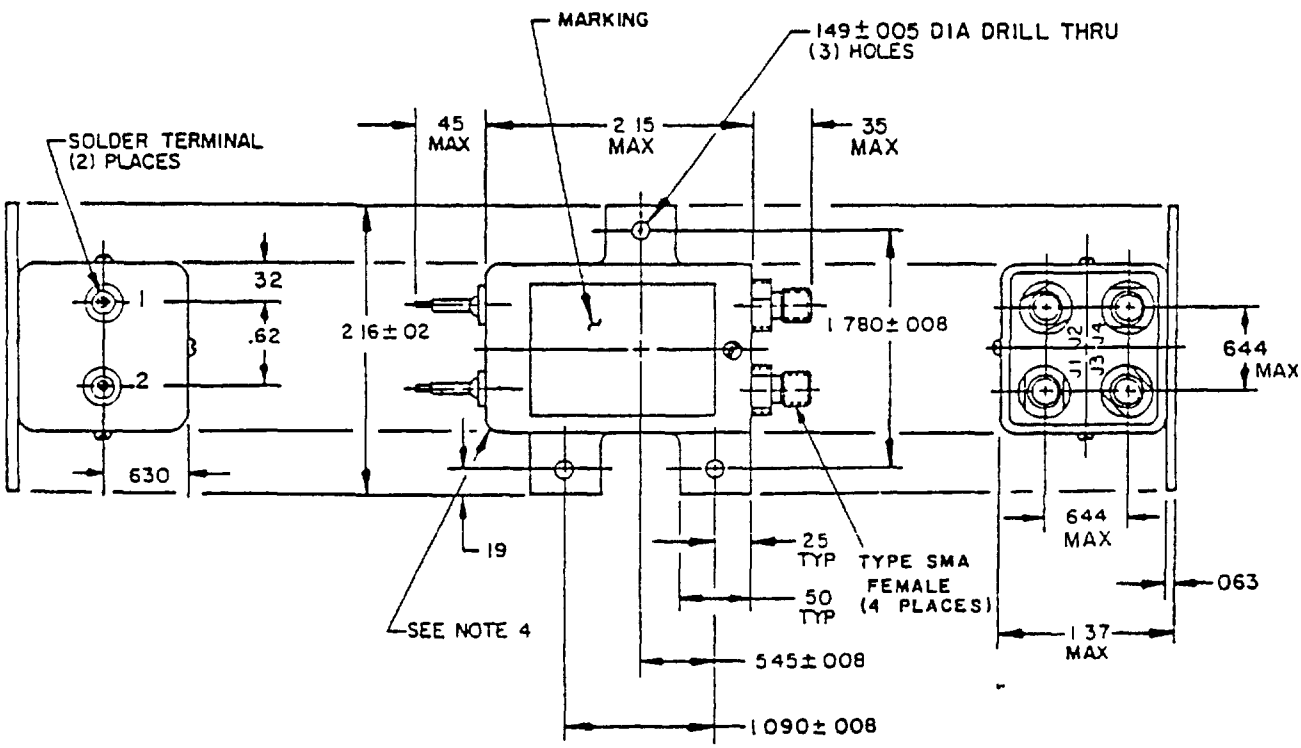
MILITARY SPECIFICATION SHEET

Ⓒ SWITCHES, RADIO FREQUENCY
TRANSMISSION LINE (COAXIAL) (ELECTRICALLY OPERATED)
CLASS 5, TR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-3928.

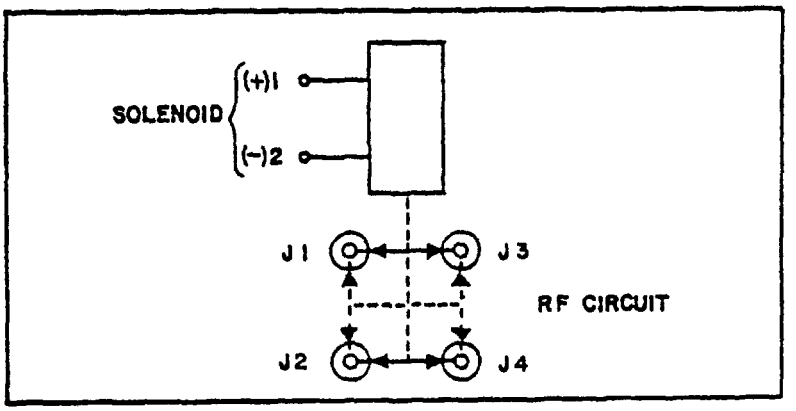


Ⓒ FIGURE 1. Part numbers M3928/19-02, M3928/19-08, and M3928/19-13

Ⓒ denotes changes

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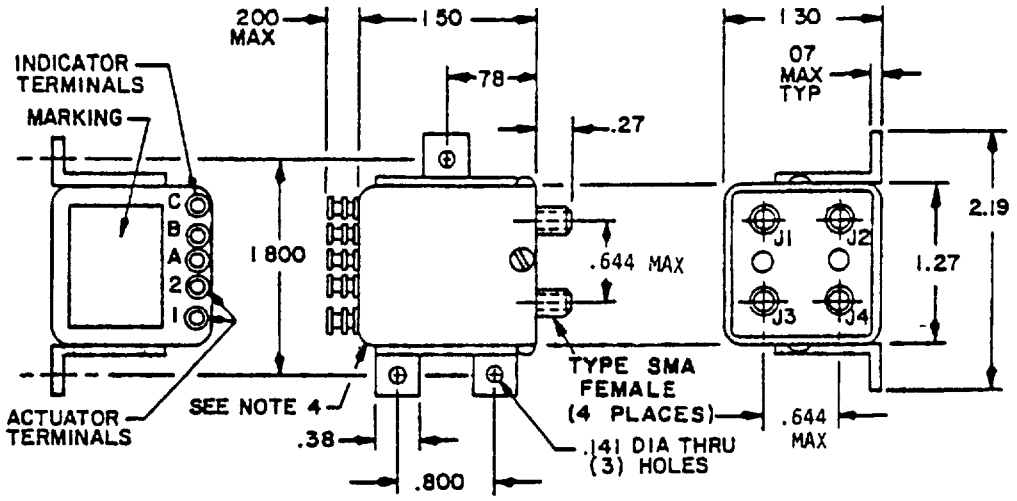
POS 1. DEENERGIZED OR FAILSAFE POSITION, J1-J3 AND J2-J4.
POS 2. ENERGIZED POSITION, J1-J2 AND J3-J4.
SCHEMATIC (SHOWN IN POSITION 1).

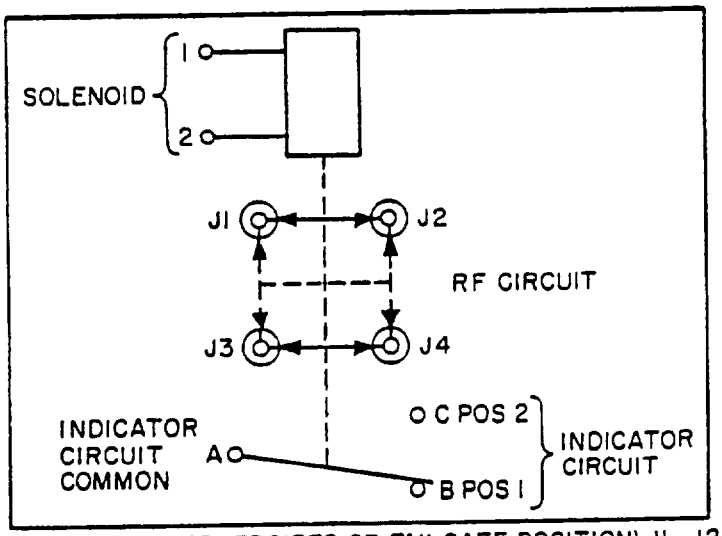
Inches	mm	Inches	mm
.005	0.13	.50	12.7
.008	0.20	.545	13.84
.02	0.5	.62	15.8
.063	1.60	.630	16.00
.149	3.78	.644	16.36
.19	4.8	1.090	27.39
.25	6.4	1.37	34.8
.32	8.1	1.780	45.21
.35	8.9	2.15	54.6
.45	11.4	2.16	54.9

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

© FIGURE 1. Part numbers M3928/19-02, M3928/19-08, and M3928/19-13 - Continued.

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POSITION 1.(DEENERGIZED OR FAILSAFE POSITION) J1 - J2 AND J3 - J4.
POSITION 2.(ENERGIZED POSITION) J1 - J3 AND J2 - J4.

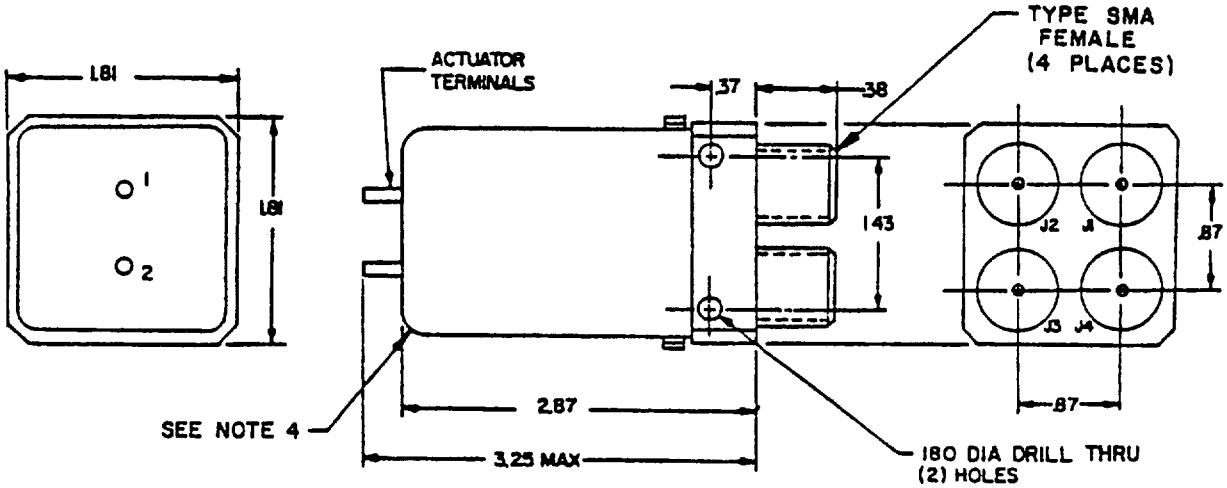
SCHEMATIC (SHOWN IN POSITION 1)

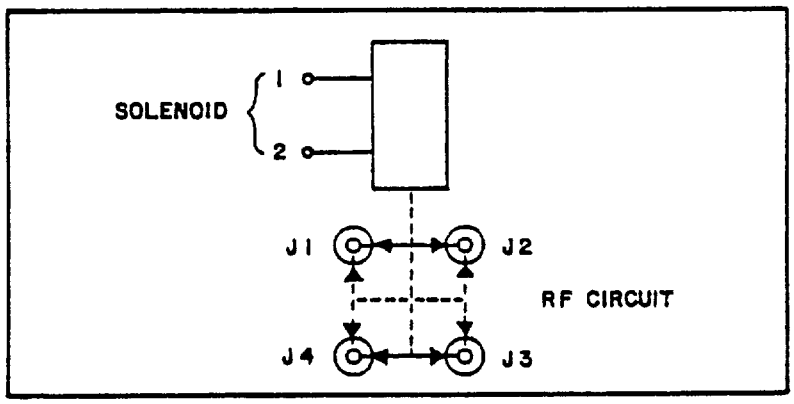
Inches	mm
0.07	1.8
0.141	3.58
0.200	5.08
0.27	6.9
0.38	9.7
0.640	16.26
0.78	19.8
0.800	20.32
1.27	32.3
1.30	33.0
1.50	38.1
1.800	45.72
2.19	55.6

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

FIGURE 2. Part number M3928/19-03.

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Inches	mm
.37	9.4
.38	9.6
.87	22.1
1.43	36.3
1.81	46.0
2.87	72.9
3.25	82.6

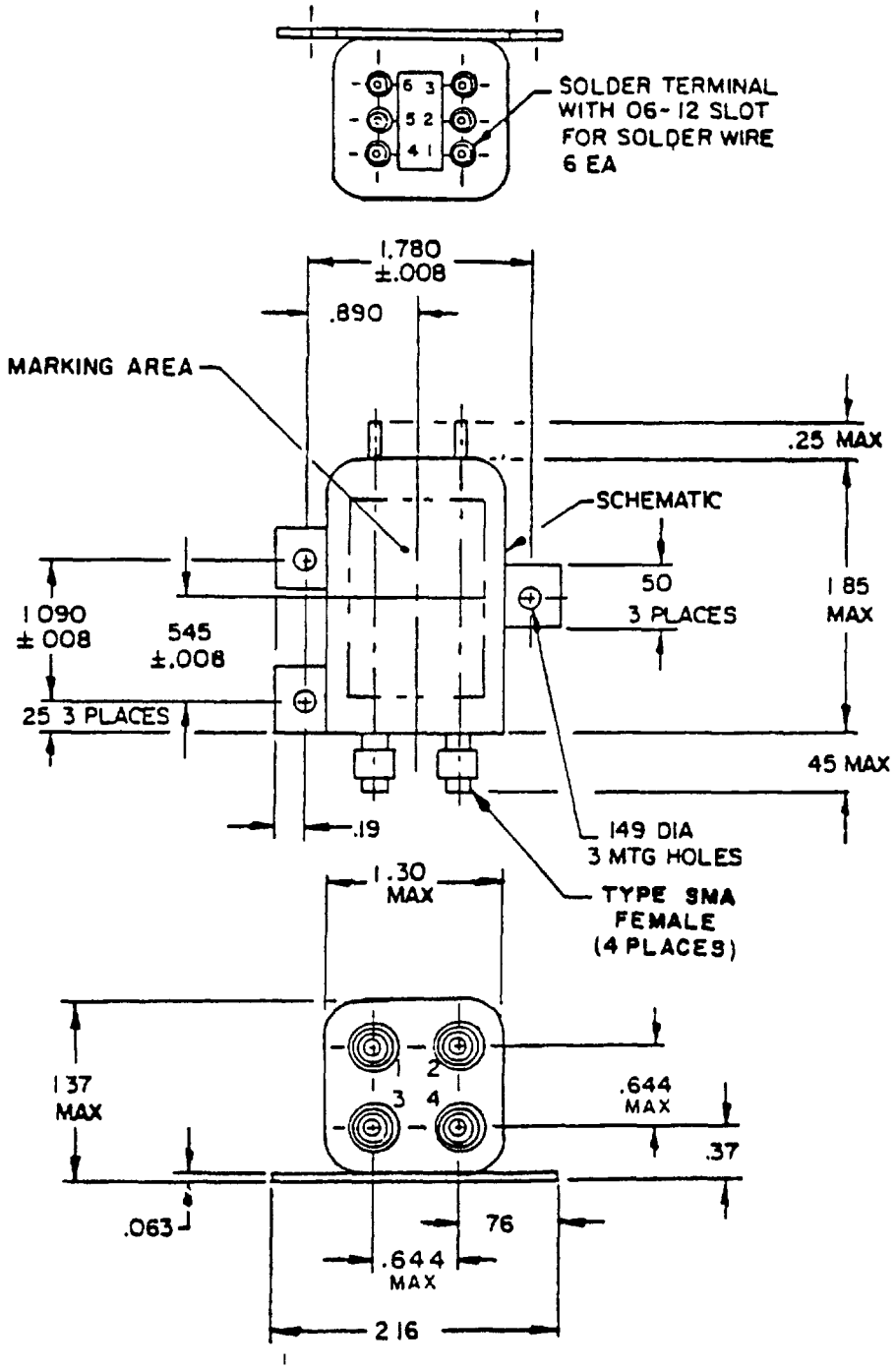
POS 1. DEENERGIZED OR FAILSAFE POSITION, J1-J2 AND J3-J4.
POS 2. ENERGIZED POSITION, J1-J4 AND J2-J3.

SCHEMATIC (SHOWN IN POSITION 1).

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

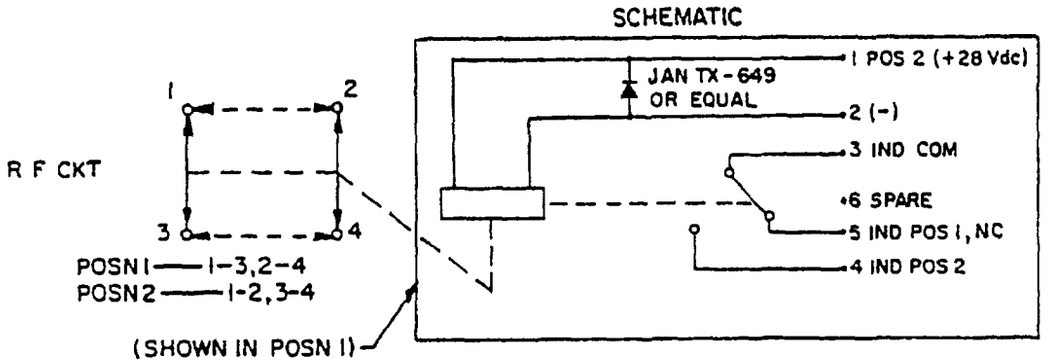
© FIGURE 3. Part number M3928/19-04.

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© FIGURE 4. Part numbers M3928/19-05, M3928/19-09, and M3928/19-14.

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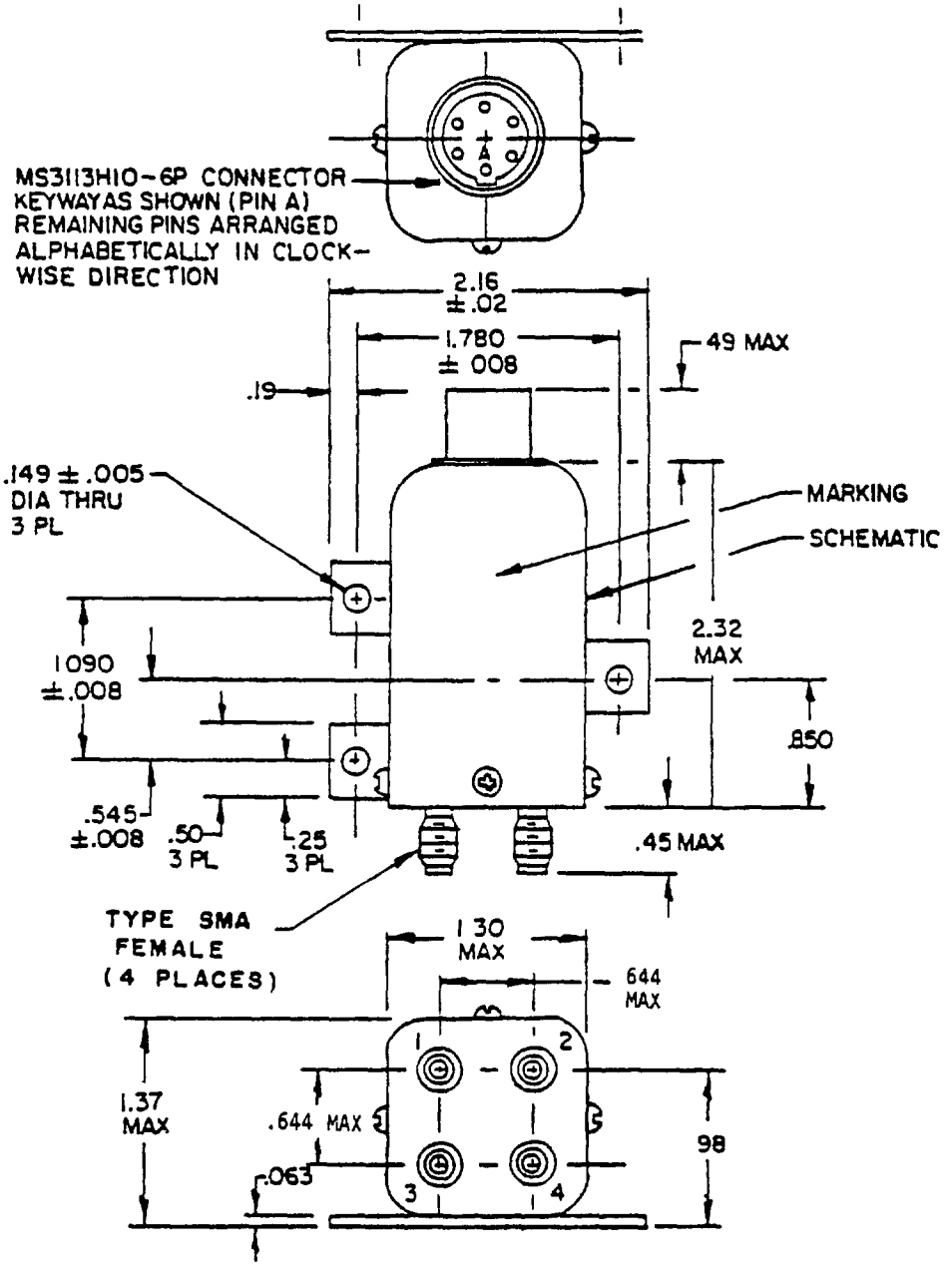


Inches	mm	Inches	mm
.008	0.20	.545	13.84
.03	0.8	.644	16.36
.06	1.5	.76	19.3
.063	1.6	.890	22.61
.12	3.1	1.090	27.69
.149	3.78	1.30	33.0
.19	4.8	1.37	34.8
.25	6.4	1.780	45.21
.37	9.4	1.85	47.0
.45	11.4	2.16	54.9
.50	12.7		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

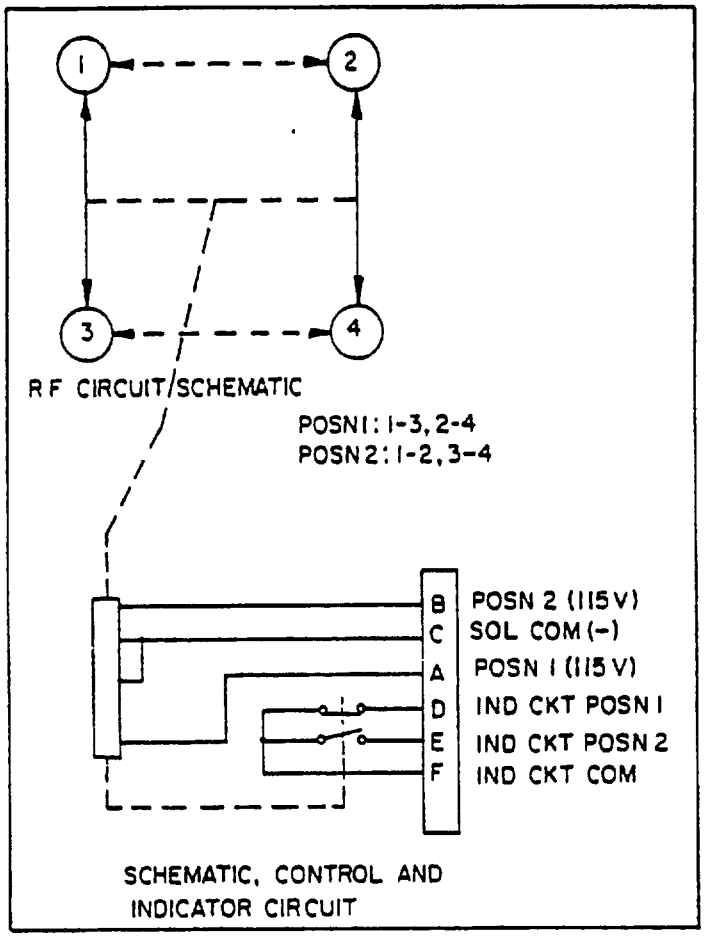
© FIGURE 4. Part numbers M3928/19-05, M3928/19-09, and M3928/19-14 - Continued.

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© FIGURE 5. Part number M3928/19-06.

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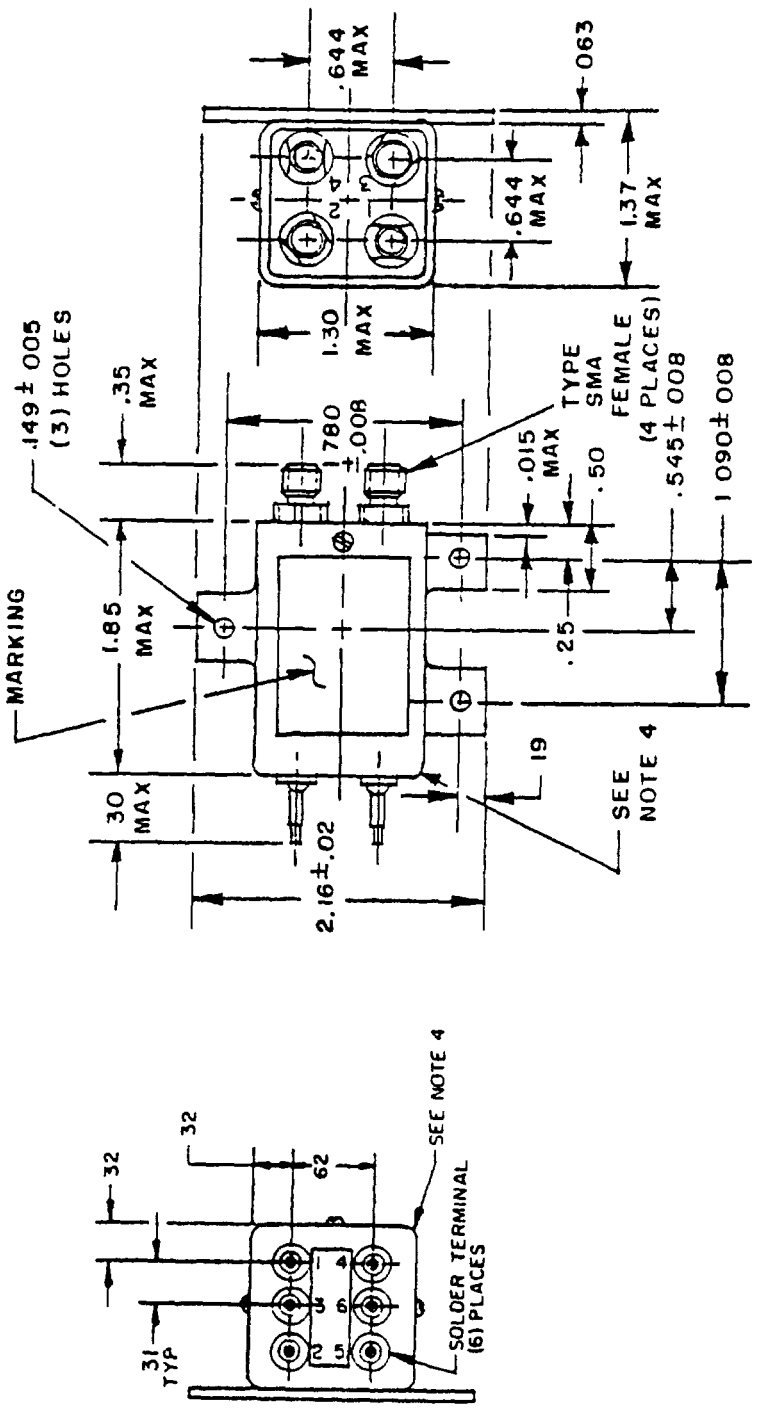


Inches	mm	Inches	mm
.005	0.13	.545	13.84
.008	0.20	.636	16.15
.02	0.5	.850	21.59
.063	1.60	.98	24.9
.19	4.8	1.090	27.69
.149	3.78	1.30	33.0
.25	6.4	1.37	34.8
.45	11.4	1.780	45.21
.49	12.5	2.16	54.9
.50	12.7	2.32	58.9

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

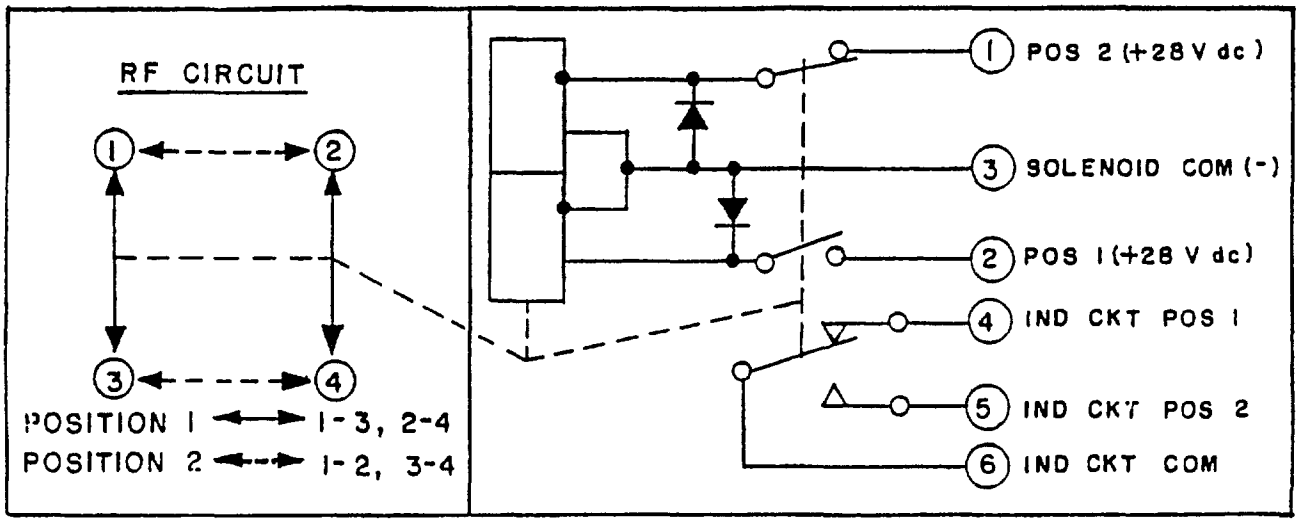
© FIGURE 5. Part number M3928/19-06 - Continued.

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© FIGURE 6. Part number M3928/19-07.

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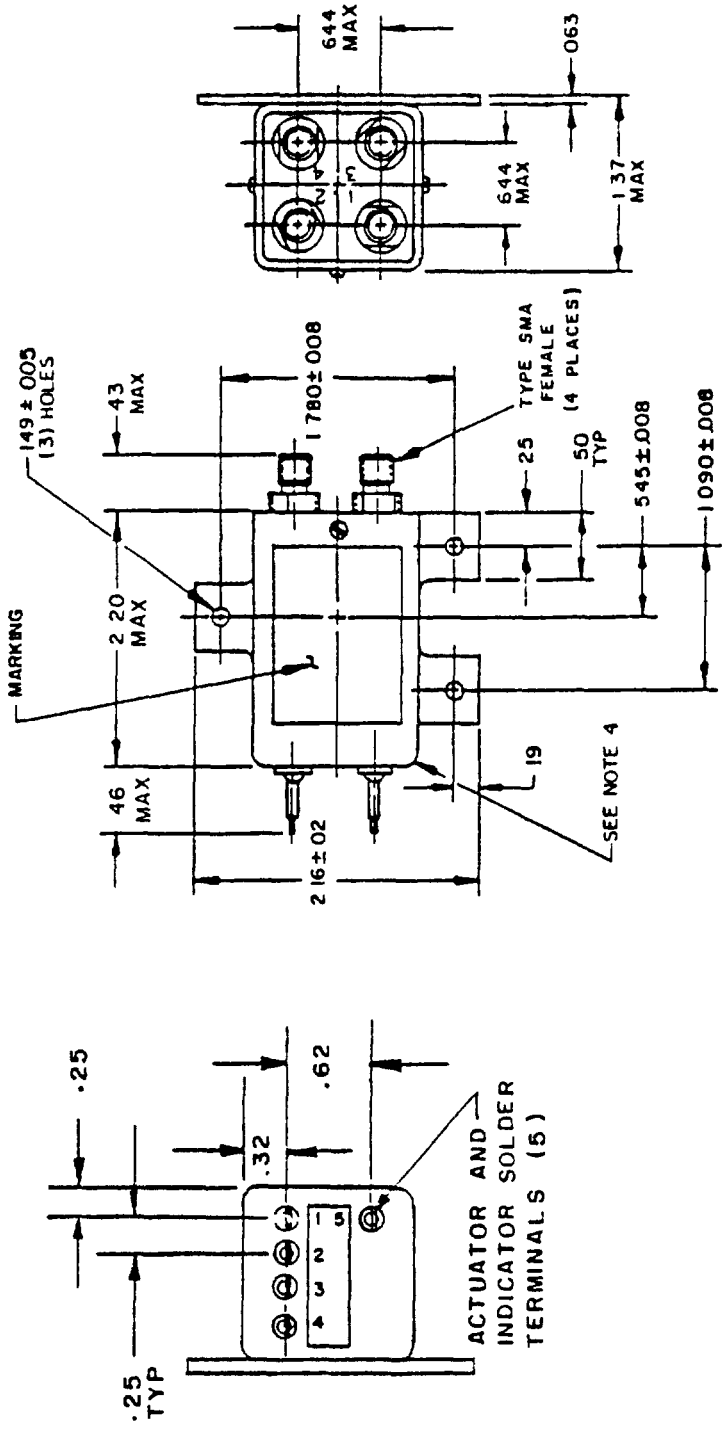
SCHEMATIC (SHOWN IN POS 1)

Inches	mm	Inches	mm
.005	0.13	.35	8.9
.008	0.20	.50	12.7
.015	0.38	.545	13.84
.02	0.5	.62	15.8
.063	1.60	.644	16.36
.149	3.78	1.090	27.69
.19	4.8	1.30	33.0
.25	6.4	1.37	34.8
.30	7.6	1.780	45.21
.31	7.9	1.85	47.0
.32	8.1	2.16	54.9

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

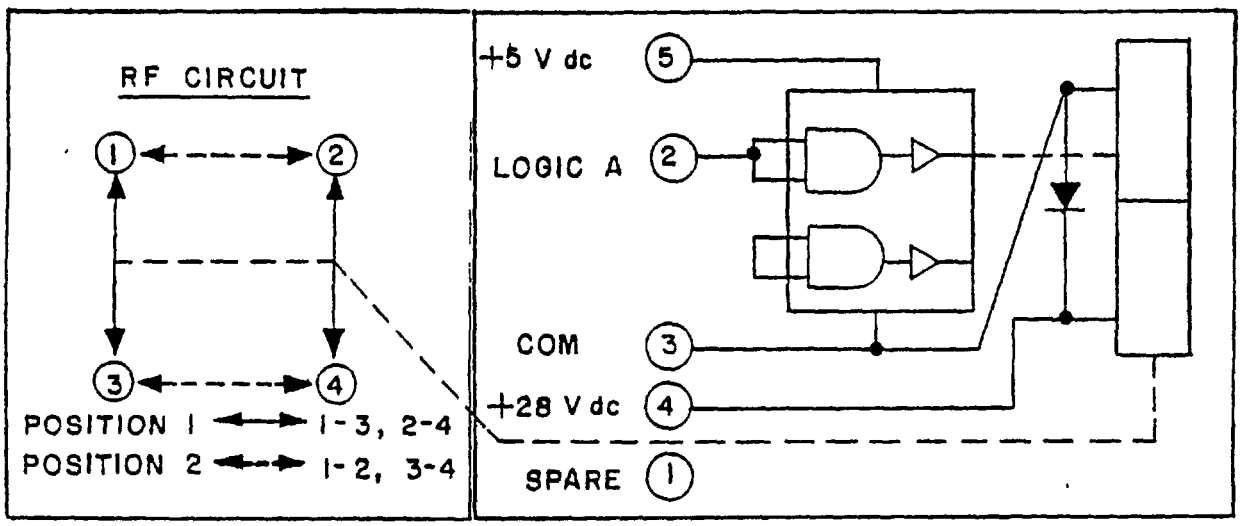
© FIGURE 6. Part number M3928/19-07 - Continued.

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© FIGURE 7. Part number M3928/19-10.

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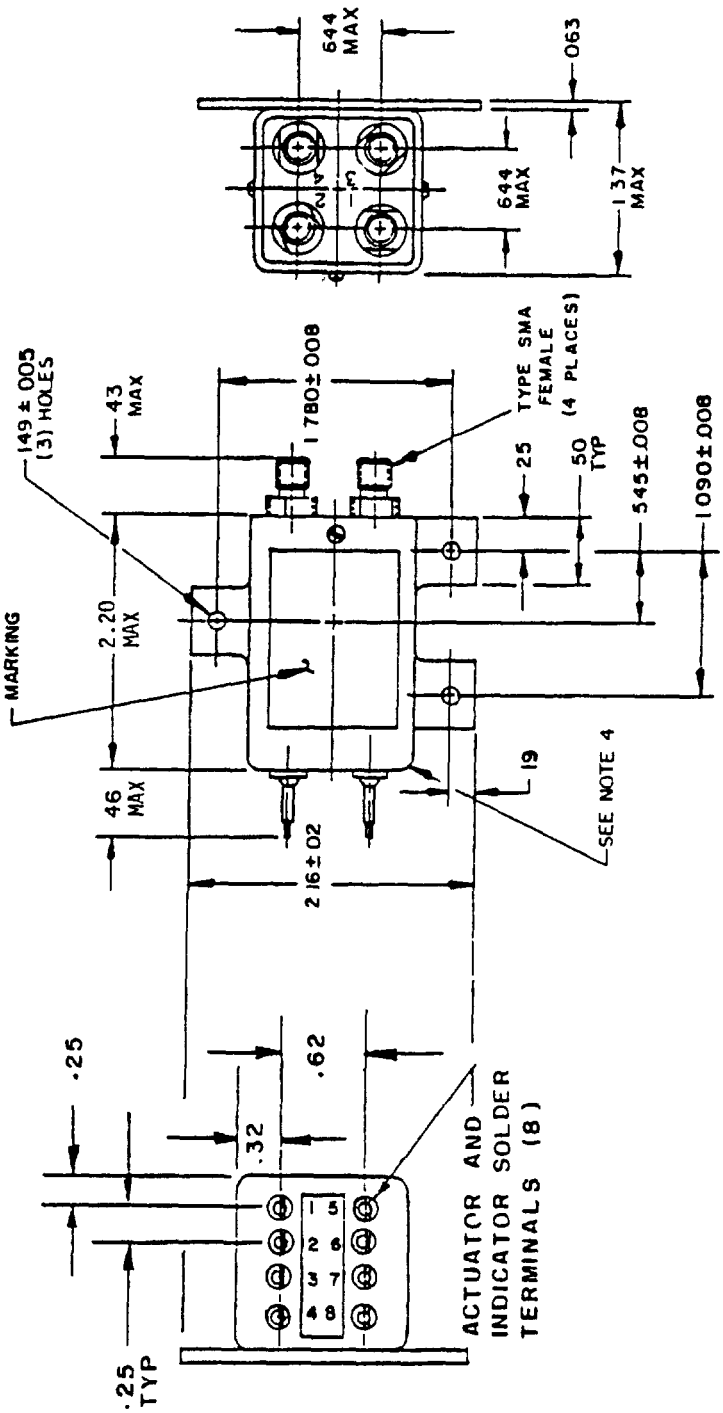
Logic truth table		
RF position	Logic signal A	
Pos 1	0	
Pos 2	1	

Inches	mm
.005	0.13
.008	0.20
.02	0.5
.063	1.60
.149	3.78
.19	4.8
.25	6.4
.32	8.1
.43	10.9
.46	11.7
.50	12.7
.545	13.84
.62	15.8
.644	16.36
1.090	27.69
1.37	34.8
1.780	45.21
2.15	54.6
2.16	54.9

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

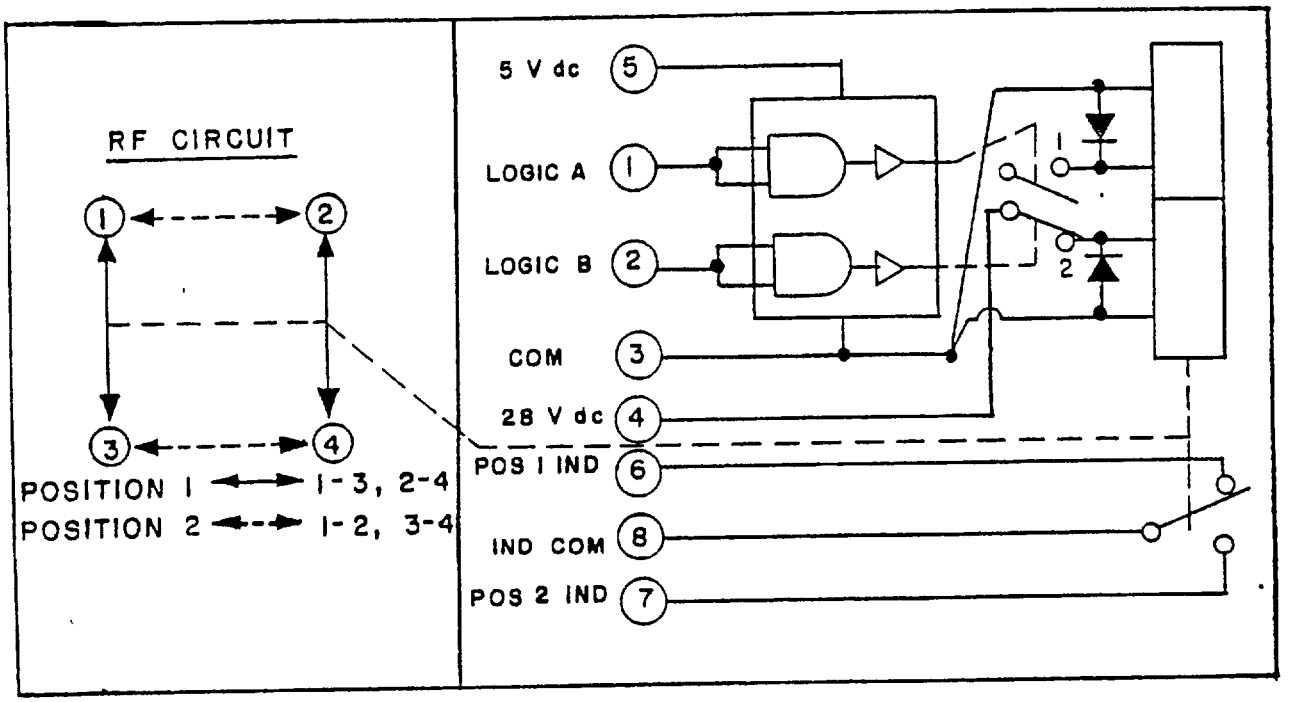
© FIGURE 7 Part number M3928/19-10 - Continued.

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© FIGURE 8. Part number M3928/19-11.

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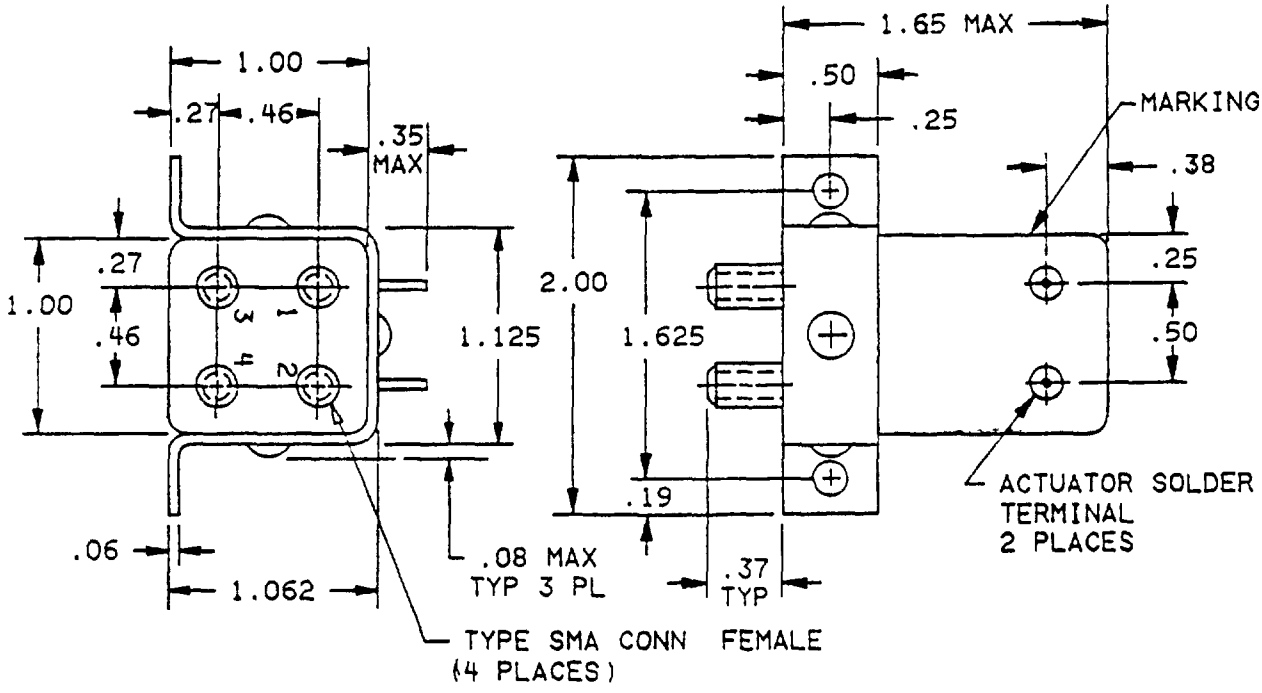
Logic truth table		
RF position	Logic signal A	B
Pos 1	1	0
Pos 2	0	1

Inches	mm
.005	0.13
.008	0.20
.02	0.5
.063	1.60
.19	4.8
.25	6.4
.32	8.1
.46	11.7
.50	12.7
.545	13.84
.644	16.36
1.090	27.69
1.37	34.8
2.15	54.6
2.16	54.9

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

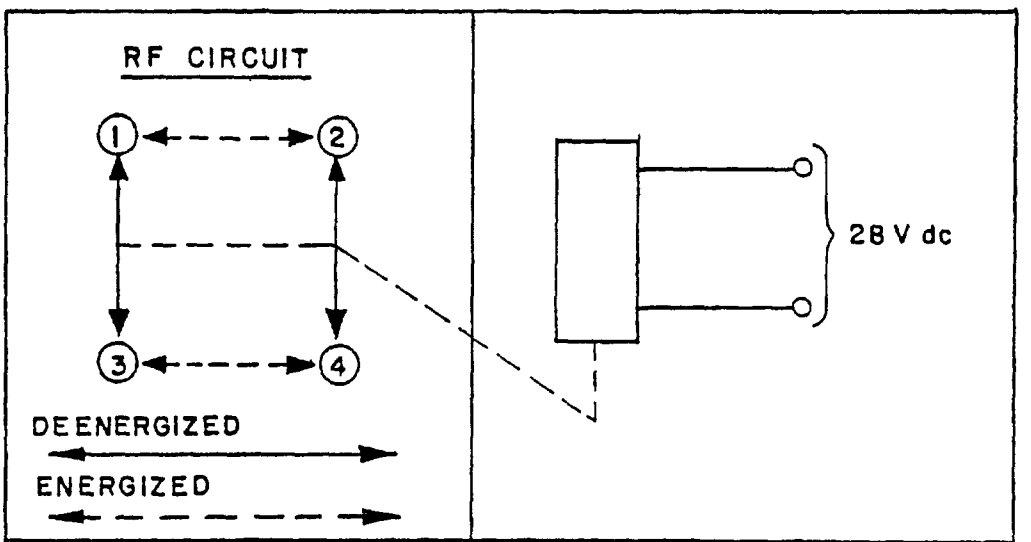
© FIGURE 8. Part number M3928/19-11 - Continued.

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© FIGURE 9 Part number M3928/19-12.

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RF CONNECTIONS SHOWN IN FAILSAFE POSITION
(DEENERGIZED)

Inches	mm
.06	0.15
.08	0.2
.156	3.96
.19	4.8
.25	6.4
.27	6.9
.33	8.4
.37	9.4
.38	9.7
.45	11.4
.46	11.7
.50	12.7
1.00	25.4
1.125	28.58
1.625	41.30
1.65	41.9
2.00	50.8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

© FIGURE 9. Part number M3928/19-12 - Continued.

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TABLE I. Electrical and performance characteristics 1/

Part no M3928/19	Figure no	Housing	Frequency range dc to GHz	VSWR	Insertion loss (dB)	Isolation (dB)	Switching time (ms)	Position indication circuit and rating	Life cycles x 1000	Fail-safe or latching	Operating current (A) 1/	Holding current (A) 1/	Nominal operating voltage	Pickup voltage	Dropout voltage	Power and indicator connector	Weight (oz)
				Max	Max	Min	Max				Max	Max					Max
02N,S	2	E	12.4	1.5:1	0.5	50	20	None	100	F	1.3	35	28 V dc	18 V dc	5 V dc	Solder terminal	6
03N,S	3	E	18	1.5:1	0.5	60	20	Yes	100	F	.175	.06	28 V dc	20 V dc	5 V dc	Solder terminal	10
4/ 04N,S	4	I	12.4	1.5:1	0.5	60	50	None	1000	F	.30	30	28 V dc	20 V dc	2 V dc	Solder terminal	14
3/ 05N,S	5	I	12.4	1.5:1	0.5	60	20	Yes	100	F	2.0	12	28 V dc	20 V dc	10 V dc	Solder terminal	4
5/ 06N,S	6	I	18	1.5:1	0.5	60	10	Yes	1000	L	.030	0	115 V ac	N/A	N/A	N/A	2.5
07N,S	7	E	18	1.5:1	0.5	60	20	Yes	100	L	.75	0	28 V dc	20 V dc	---	Solder terminal	6
08N,S	2	E	18	1.5:1	0.5	60	20	None	100	F	1.3	35	28 V dc	18 V dc	5 V dc	Solder terminal	6
09N,S	5	E	18	1.5:1	0.5	60	20	Yes	100	F	2.0	12	28 V dc	20 V dc	10 V dc	Solder terminal	4
6/ 10N,S	8	E	18	1.5:1	0.5	60	20	None	100	F	1.3	35	28 V dc +5 V dc 7/	18 V dc	5 V dc	Solder terminal	7
6/ 11N,S	9	E	18	1.5:1	0.5	60	20	Yes	100	L	1.0	0	28 V dc +5 V dc 7/	20 V dc	--	Solder terminal	7
12N,S	10	E	18	1.5:1	0.5	60	20	None	100	F	1.0	2	28 V dc	20 V dc	16 V dc	Solder terminal	4
13N,S	2	E	20	1.8:1 1.5:1@ 18 GHz	0.7 0.5@ 18 GHz	40 60@ 18 GHz	20	None	100	F	1.3	95	28 V dc	18 V dc	5 V dc	Solder terminal	6
			8	1.3:1	0.3	70											
14N,S	5	E	20	1.8:1 1.5:1@ 18 GHz	0.7 0.5@ 18 GHz	40 60@ 18 GHz	20	Yes	100	F	2.0	12	28 V dc	20 V dc	5 V dc	Solder terminal	6

1/ M3928/19-01 has been deleted.

2/ At nominal V dc and 20°C.

3/ Transient interference (RFI).

4/ Explosion proof. 25 watts at 1 GHz into a 50-ohm load.

5/ Switch does not provide self cutoff of operating voltage after switching.

6/ Logic voltage: Logic 0 - 0 to 0.4 V dc
Logic 1 - 2.4 to 5.5 V dc

7/ +5 voltage deviation ± 0.5 V dc.

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REQUIREMENTS:

© Dimensions and configurations: See figures 1 through 9.

RF connectors: Female connectors (4 places) shall meet the requirements of MIL-C-39012 and shall mate with SMA type male connectors per MIL-C-39012/55.

Electrical and performance characteristics: See table I.

RF power handling capability (average): 25 watts (minimum).

Nominal impedance: 50 ohms.

Termination: Open.

Vibration: Method I.

Operating temperature: -55°C to +85°C.

Part number: M3928/19- (and dash number from table I).

Substitutability: Switches covered by this specification sheet are substitutable for the manufacturer's or original equipment manufacturer's (OEM) part numbers specified in table II. This information in no way implies that the manufacturer's or OEM part number or NSN is suitable as a substitute for the military part number.

© TABLE II. Substitution information.

Part number M3928/19-	OEM or manufacturer's code	OEM or manufacturer's part number	NSN 5985-
02	82152	710C70100	125-9895
02	07397	31-815065-1	125-9895
02	02660	303-11022-53	---
03	07397	20-808533-1	---
03	11312	CS-37S1C-1	---
04	97942	580R605H01	118-4719
04	96341	MA7525TMD	118-4719
04	02660	303-11016-3	---
05	49956	584071-1	625-9681
05	82152	710C71400	---
06	07397	20-807756	5865-00-204-4340
06	24022	CS37S7C-2	---
07	82152	700C70200	4920-01-022-8562
08	82152	710C70100	---
09	82152	710C71400	---
10	82152	710C70100-30	---
11	82152	700C70200-30	---
12	82152	715C70100	525-3799
13	82152	710C70100	---
14	82152	710C71400	---

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - MI
Navy - OS
Air Force - 11, 99
DLA - ES

User activities:
Army - AV
Navy - AS, MC, SH
Air Force - 19

Preparing activity.
Navy - EC

Agent:
DLA - ES

(Project 5985-1016-11)

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